MIL-P-70486 (AR)
AMENDMENT 1
13 August 1990

MILITARY SPECIFICATION

PROCESSOR, SIGNAL, FIRE AND FLIGHT AIR DATA SUBSYSTEM, HELICOPTER ARMAMENT: M143

This amendment forms a part of Military Specification MIL-P-70486 (AR), dated 30 June 1983, and is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

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2.1, Under Military Specifications: Delete "MIL-S-45743 Soldering, Manual Type, High Reliability, Electrical, Electronic, Instrument, Communication and Radar for Aerospace and Control Systems, Procedure for" and "MIL-S-46844 Solder Bath, Soldering and Printed Wiring Assemblies, Automatic Machine Type" and substitute "MIL-STD-2000 Standard Requirements For Soldered Electrical and Electrical Assemblies".

Under Military Standards: Delete "MIL-STD-810C" and substitute "MIL-STD-810".

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3.4.5.1.3: Delete in its entirety and substitute:

"3.4.5.1.3 Humidity. Operation while being subjected to MIL-STD-810, Method 520.1, Procedure III.".

3.4.5.1.4: Delete in its entirety and substitute:

"3.4.5.1.4 Sand and dust. Operation after being subjected to MIL-STD-810, Method 510.3, Procedure II.".

3.4.5.1.5: Delete "Method 509.1" and substitute "Method 509.3".

3.4.5.1.6: Delete "Method 508.1" and substitute "Method 508.4".

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3.4.5.1.7: Delete "Method 511.1" and substitute "Method 511.3".

3.4.5.1.8: Delete in its entirety and substitute:

"3.4.5.1.8 Rain. Operation after being subjected to MIL-STD-810, Method 506.3, Procedure I. The EPU shall be operated during the last ten minutes of the thirty minute period.".

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3.4.5.3.1: Delete in its entirety and substitute:

"3.4.5.3.1 Vibration. The EPU shall not sustain damage or subsequently fail to provide specified performance when subjected to the vibration environment defined by Figure 5.".

3.4.5.3.2: Delete in its entirety and substitute:

"3.4.5.3.2 Shock. The EPU shall meet the shock requirements of MIL-STD-810, Method 516.4, Procedure I and Method 516.4, Procedure V.".

3.4.6: Delete in its entirety and substitute:

"3.4.6 Transportability. The EPU shall withstand vibration environments encountered during transportation as defined in Fig. 11.".

3.4.7.1: Delete "MIL-S-45743 or MIL-S-46844" and substitute "MIL-STD-2000 General Requirements with Task F".

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TABLE III: Delete in its entirety and substitute:

"TABLE III. First article tests

Item	Characteristic	Requirement	Test Procedure
1	Natural environment	3.4.5.1	4.6.4.1
2	Altitude	3.4.5.1.1	4.6.4.1.1
3	Temperature	3.4.5.1.2	4.6.4.1.1
4	Humidity	3.4.5.1.3	4.6.4.1.1
5	Sand and dust	3.4.5.1.4	4.6.4.1.2
6	Salt atmosphere	3.4.5.1.5	4.6.4.1.3
7	Fungus	3.4.5.1.6	4.6.4.1.4
8	Explosive atmosphere	3.4.5.1.7	4.6.4.1.5
9	Rain	3.4.5.1.8	4.6.4.1.6
10	Interchangeability	3.4.9	Applicable drawings,visual, MIL-I-8500 and MIL-STD-100
11	Service environment	3.4.5.2	4.6.4.2
12	Bench handling	3.4.5.2.1	4.6.4.2.1
13	Induced environment	3.4.5.3	4.6.4.3
14	Vibration	3.4.5.3.1	4.6.4.3.1
15	Shock	3.4.5.3.2	4.6.4.3.2
16	Transportability	3.4.6	4.6.4.4
17	Electromagnetic radiation".	3.4.8	4.6.4.5

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4.6.4.2: Delete in its entirety and substitute:

"4.6.4.1.1 Temperature-altitude-humidity. The EPU shall be subjected to the combined Temperature-altitude-humidity test of MIL-STD-810, Method 520.1, Procedure III, Figure 520.1-5, for ten test cycles. The EPU shall be tested to a maximum operating, nonoperating, and storage temperature of 71 degrees C and a minimum operating, nonoperating and storage temperature of -54 degrees C, a maximum flight altitude of 20,000 ft. and a maximum humidity of 94% ±4% at 30 degrees C. At the conclusion of this test, Subject the EPU to the performance tests of Table IV. The EPU shall be exposed to the Temperature-altitude-humidity test before the Sand and dust and Explosive atmosphere tests and after the Bench handling, Crash safety, and Vibration tests.".

4.6.4.3: Delete in its entirety.

4.6.4.4: Delete in its entirety.

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4.6.4.5: Delete in its entirety and substitute:

"4.6.4.1.2 Sand and dust. The EPU shall be subjected to test Method 510.3, Procedure II, of MIL-STD-810. The test chamber shall be set to the high operating temperature of the EPU of 71 degrees C, the air velocity should be of the range 18 to 29 m/s, and the sand concentration shall be 2.2 ± 0.5 g/m^3. At the conclusion of this test, subject the EPU to the performance tests of Table IV. The EPU shall be subjected to the Sand and dust test after being subjected to the Fungus, Temperature-altitude-humidity, and Salt atmosphere tests.".

4.6.4.6: Delete in its entirety and substitute:

"4.6.4.1.3 Salt atmosphere. The EPU shall be subjected to the test method 509.3 of MIL-STD-810. Operation of the EPU shall not be required and failure criteria shall be limited to corrosion of finishes and metals only. Such corrosion shall be defined as visible degradation of the equipment surfaces that can be attributed to flaky, pitted, blistered, or otherwise loosened finish or metal surface. The EPU shall be subjected to the salt atmosphere test prior to being subjected to the Sand and dust test.".

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4.6.4.7: Delete in its entirety and substitute:

"4.6.4.1.4 Fungus. The EPU shall be subjected to the test of method 508.4 of MIL-STD-810. The test shall have a duration of 28 days. Gasket sealed or other non-hermetic sealed test items shall be opened during test exposure and all internal and external surfaces shall be sprayed with spore suspension. Operation of the equipment is not required, but examination of the concentration after exposure period shall reveal no evidence of viable fungus on any surfaces. The EPU shall be subjected to the fungus test prior to being subjected to the salt atmosphere test and the sand and dust test.".

4.6.4.8: Delete in its entirety and substitute:

"4.6.4.1.5 Explosive atmosphere. The EPU shall be subjected to test Method 511.3, Procedure I, of MIL-STD-810. The test shall be limited to a maximum altitude of 20,000 feet. At the conclusion of this test, subject the system to the performance tests of table IV. The EPU shall be subjected to the explosive atmosphere test after being subjected to the vibration test and the temperature-altitude-humidity test.".

4.6.4.9: Delete in its entirety and substitute:

"4.6.4.1.6 Rain. The EPU shall be subjected to test Method 506.3, Procedure I, of MIL-STD-810, with a rainfall rate of 4 in./hr and a wind velocity of 40 mph. The system shall be operated during the last ten minutes of the thirty minute rain. At the conclusion of this test, subject the EPU to the performance tests of Table IV.".

4.6.4.10: Delete "4.6.4.10" and substitute "4.6.4.2".

4.6.4.10.1: Delete in its entirety and substitute:

"4.6.4.2.1 Bench Handling. The system shall be subjected to Method 516.4, Procedure VI, of MIL-STD-810. At the conclusion of this test, subject the EPU to the performance tests of Table IV. The EPU shall be subjected to this Bench handling test prior to being subjected to the Temperature-altitude-humidity test.".

4.6.4.11: Delete "4.6.4.11" and substitute "4.6.4.3".

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4.6.4.11.1: Delete in its entirety and substitute:

"4.6.4.3.1 Vibration. The EPU shall be vibrated along each axis at the sinusoidal frequencies and amplitudes depicted in Fig 5. The frequency of applied vibration shall be cycled logarithmically from 5 to 500 to 5 cycles per second for 5 cycles, each cycle having a duration of 36 minutes, totaling 3 hours testing time for each axis. At the conclusion of this test, subject the EPU to the performance test of Table IV. The EPU shall be subjected to the vibration test prior to being subjected to any other environmental test.".

4.6.4.11.2: Delete in its entirety and substitute:

"4.6.4.3.2 Shock. The EPU shall be subjected to:

a. Method 516.4, Procedure I, of MIL-STD-810, for flight vehicle equipment. The test 'g' value shall be within ±10% when measured with a 0.2 to 250 Hz filter. The pulse test level and duration are defined in paragraph 3.2.24.6.1 of MIL-E-5400.

b. Method 516.4, Procedure V, of MIL-STD-810 for flight equipment. The EPU shall be subjected to this method prior to being subjected to the Temperature-altitude-humidity test.".

4.6.4.11.3: Delete in its entirety and substitute:

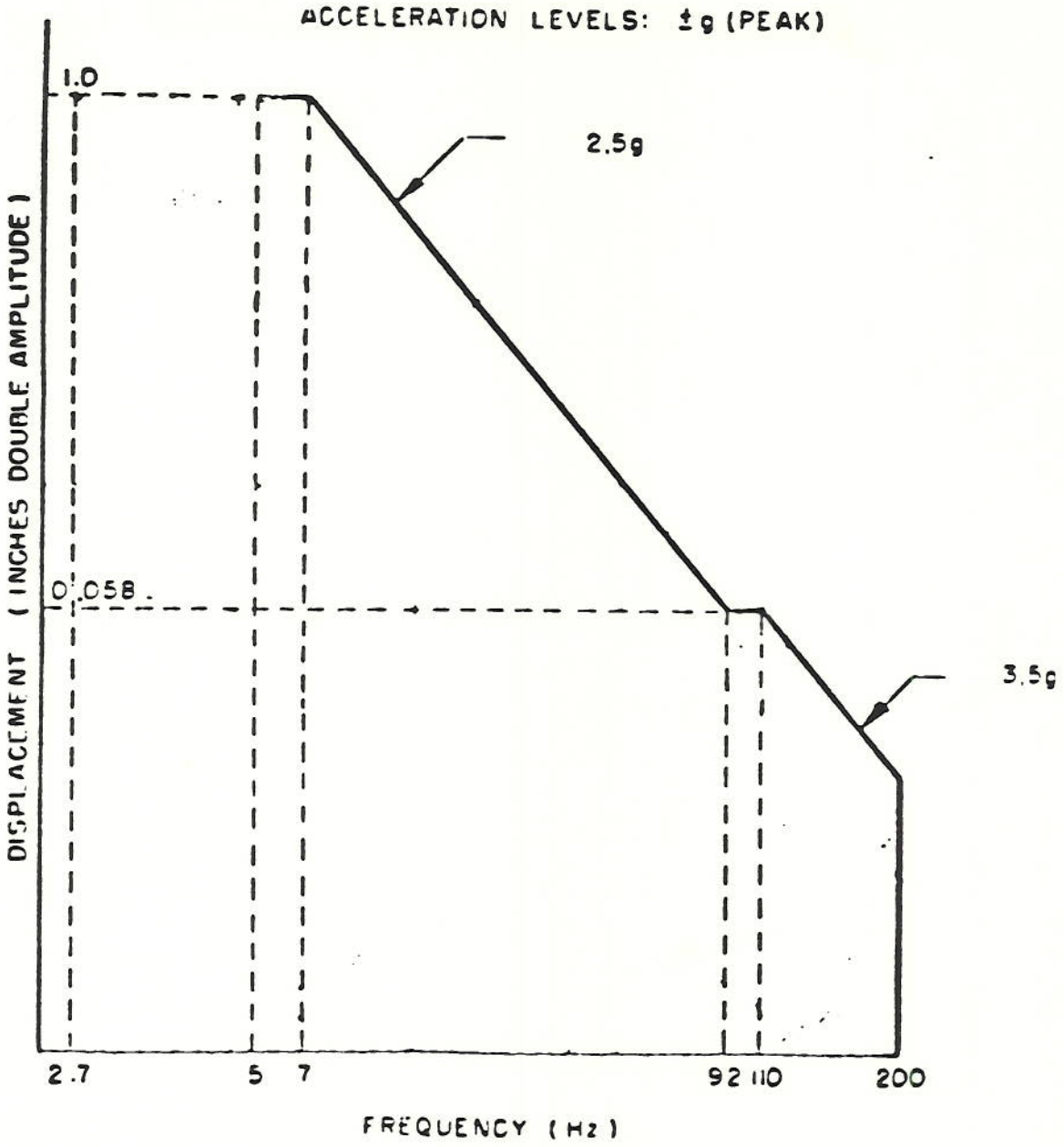
"4.6.4.4 Transportability. The EPU shall be unpacked and secured to the vibration table and shall be vibrated along each axis at the sinusoidal frequencies and amplitudes depicted in Fig. 11. The frequency of applied vibration shall be cycled logarithmically from 5 to 200 to 5 Hz, each cycle having a sweep time of 12 minutes. The system shall be exposed to the cycling for 84 minutes per axis. The EPU will not be operated during vibration. At the conclusion of this test, subject the EPU to the performance tests of Table IV.".

Add enclosed figure 11 (36A).

4.6.4.12: Delete in its entirety and substitute:

"4.6.4.5 Electromagnetic radiation. The EPU shall be installed in laboratory test fixtures in a manner that will simulate actual service usage and tested to MIL-STD-461 for the requirements of 3.4.8.".

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NOTE: ALL CURVES SHALL BE EXTENDED TO 2 Hz WHEN TEST ITEM RESONANCES BELOW 5 Hz ARE EXPECTED.

FIGURE 11 VIBRATION TEST CURVE FOR EQUIPMENT TRANSPORTED AS SECURED CARGO

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